# Comparative Security and Usability Analysis of Digital Payment Authentication Mechanisms in Nepal: A Study of eSewa, Khalti, and Mobile Banking Applications

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# Abstract

Digital payment platforms have become central to everyday financial activity in Nepal, with eSewa, Khalti, and commercial mobile banking applications collectively serving millions of users across urban and rural areas. As transaction volumes grow, so do concerns about the adequacy of security measures protecting those transactions, and whether those measures come at the cost of usability. Despite the rapid expansion of Nepal's digital payment ecosystem, no study has directly compared how these platforms implement and balance authentication security and user experience.

This study examines the authentication mechanisms of three platforms, eSewa, Khalti, and a leading commercial bank's mobile application, through a mixed-methods approach combining structured app analysis, a user perception survey, and timed performance observation. The app analysis, user survey of approximately 100 respondents drawn from urban, semi-urban, and rural backgrounds, and timed performance observation together capture authentication implementation, user awareness, trust, and usability perception.

The study is guided by the hypothesis that digital wallet applications prioritize usability over security, resulting in comparatively lower authentication standards than commercial bank apps. Findings are expected to reveal gaps in both technical implementation and user awareness, particularly around biometric feature adoption and OTP vulnerability. The study contributes the first platform-specific comparative analysis of authentication in Nepal's digital payment sector and offers targeted recommendations for developers, financial institutions, and policymakers working toward a more secure and accessible digital financial ecosystem.



**Keywords:** *digital payments, authentication, usability, security, eSewa, Khalti, mobile banking, Nepal*

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# 1. Introduction

## 1.1 Background and Context

Digital payments have transformed financial transactions in Nepal over the last decade. What was once required a trip to the bank or carrying cash now takes a few taps on a smartphone. Platforms like eSewa and Khalti have become household names, especially in the city areas, and mobile banking apps from leading commercial banks have also brought traditional banking services directly to users' phones. The COVID-19 pandemic accelerated this shift dramatically. When physical contact became risky, contactless payments stopped being a convenience and became a necessity.

The roots of this transformation go back further. Nepal's digital banking era began with Nabil Bank introducing credit cards in the 1990s, but the real turning point came in 2004 when Laxmi Bank launched the country's first SMS banking service [1]. That milestone marked the beginning of a gradual shift away from physical branch dependency toward digital self-service. From SMS alerts to full mobile wallets, the journey has remained steady.

Globally, this growth is not unique to Nepal. Research indicated that by 2025, roughly 25% of all global transactions would be handled through instant digital payments or e-money [2], a projection that has largely materialized. South Asian countries, including Nepal, have been part of this wave, though not without friction. Poor internet infrastructure, limited digital literacy in rural areas, and concerns about online fraud continue to slow adoption in many parts of the country [2]. Still, the numbers move in one direction, upward.

What makes Nepal's case particularly interesting is how quickly user behavior has changed despite these barriers. Users who had no prior experience with mobile banking five years ago are now scanning QR codes at local stores and even sending dakshina during Dashain and Tihar festivals. Brand trust and ease of use appear to drive adoption more significantly than technical specifications [3].

## 1.2 Problem Statement

Despite this rapid growth, there are real and underexplored concerns about how secure these platforms actually are, and whether users even understand what security

features they are using. Security threats in Nepal's e-payment sector have been growing in direct proportion to the volume of transactions [4]. Identity theft, phishing, and account takeovers are not abstract risks anymore. These threats increasingly affect actual users.

At the same time, making an app too secure can backfire. Research shows that users often perceive a trade-off between the convenience of e-payments and the risk of personal data misuse [5]. Excessive authentication steps often lead to user abandonment [6]. This tension between security and usability is the core challenge that app developers and policymakers in Nepal need to address. Yet no comparative study has specifically examined how eSewa, Khalti, and mobile banking apps handle this balance [7].

### 1.3 Research Objectives

This study aims to fill that gap. Specifically, the objectives are:

1. To identify and compare the authentication mechanisms used by eSewa, Khalti, and a leading commercial bank's mobile banking application.
2. To evaluate user perception of security and trust across the three platforms through survey data collected from Nepali users.
3. To analyze the usability trade-offs associated with different authentication methods and identify what balance works best in the Nepali context.
4. To provide practical recommendations for improving both security and user experience in Nepal's digital payment platforms.

This study is guided by the following hypothesis:

*H1: Digital wallet applications in Nepal prioritize usability over security, resulting in lower authentication standards compared to commercial mobile banking applications.*

### 1.4 Scope and Significance

This research focuses specifically on the authentication layer of three platforms, eSewa, Khalti, and the mobile banking application of a leading Nepali commercial bank. The scope does not extend to backend infrastructure, blockchain implementation, or legal and regulatory compliance. The focus is deliberately narrow: how do users log in, verify transactions, and stay protected, and how easy or frustrating is that process?

That said, the significance goes beyond these three apps. Nepal's digital payment ecosystem is still young enough that small improvements in security design can have large effects on adoption and trust. If users don't feel safe, they won't use the apps, and the potential benefits of financial inclusion get lost. Studies consistently show that in the Nepali context, trust built on perceived security and privacy is the foundation everything else rests on [4]. By comparing three platforms side by side, this study hopes to offer insights useful not just for these specific apps but for the broader conversation about how developing economies can build secure, user-friendly digital financial systems.

Biometric methods, including fingerprint scanning, face recognition, and emerging behavioral approaches, represent a promising direction worth examining [9]. Whether Nepali users are ready for these technologies, and whether current apps are even implementing them properly, is part of what this study intends to explore.

# 2. Literature Review

## 2.1 Digital Payment Systems Overview

The global shift toward cashless transactions has been one of the defining economic trends of the past decade. As noted earlier, this shift is expected to continue, with digital payments accounting for approximately 25% of global cashless transactions. This growth has been particularly visible across Asia, where mobile-first populations and expanding smartphone penetration have created fertile ground for digital wallet adoption [5].

Nepal occupies a unique position within this regional trend. What began as a modest SMS banking experiment has evolved into a competitive market featuring dedicated digital wallets and full mobile banking applications. Today, eSewa and Khalti dominate the digital wallet space, while commercial banks offer their own mobile banking platforms, many of which are built on F1Soft infrastructure. This layered ecosystem reflects both the opportunity and the complexity of Nepal's current digital finance landscape.

However, Nepal's transition has not been smooth. As outlined in the introduction, Nepal's digital payment adoption faces several structural barriers, particularly outside urban centers [2]. This consumer behavior pattern has important implications for how security features are designed and communicated to users.

## 2.2 Authentication Mechanisms

Authentication in digital payment systems broadly falls into three categories: knowledge-based factors such as passwords and PINs, possession-based factors such as OTPs delivered via SMS, and biometric factors such as fingerprint or facial recognition [7]. In practice, most platforms combine these layers, though the depth and consistency of implementation varies considerably.

Passwords and PINs alone are no longer considered sufficient protection against modern threats. Research on behavioral biometrics highlights that static credentials are easily compromised and that more dynamic forms of authentication are needed [10]. Multi-factor authentication, which combines two or more of these categories, has emerged as the primary defense mechanism recommended across the literature [8].

In Nepal, major digital wallets and commercial bank apps currently implement biometric login alongside knowledge-based MPINs or passwords, with SMS OTP reserved for high-risk actions such as large fund transfers. Preliminary examination of these platforms reveals that biometric login, whether fingerprint or face recognition, works through the phone's native security system rather than the app itself. The app simply asks the device to verify the owner, and the device uses whatever biometric the user has set up. Face recognition demonstrates consistent reliability on iPhone X and later models, whereas Android device support varies significantly. Many budget Android phones use 2D camera-based face unlock, which can be bypassed using a photograph. Nepal Rastra Bank guidelines push platforms toward stronger authentication standards, leading many apps to effectively disable face login on such devices, defaulting to fingerprint or PIN. This means biometric support is not a uniform experience across Nepal's user base.

Multimodal biometric systems that combine fingerprint and facial recognition have demonstrated over 94% improvement in reducing false acceptance rates compared to single-trait systems [9]. Behavioral biometrics, which authenticate users continuously through typing patterns and touch dynamics, represent an even more advanced direction [10]. None of the platforms currently under study implement anything in this category, which is a notable gap given where global research is heading.

### 2.3 Security Frameworks and Standards

From a framework perspective, multi-factor authentication is consistently identified as the foundational requirement for securing digital financial transactions [8]. Beyond static authentication, real-time fraud detection systems are increasingly considered essential as transaction volumes grow and attack patterns become more sophisticated. The literature points toward AI-driven personalized security as the next step, one that adapts to individual user behavior rather than applying uniform rules to everyone.

Insufficient authentication remains the leading cause of information attacks in cloud-based banking environments [7]. This is a concern directly relevant to Nepal, where platforms handle sensitive financial data on infrastructure that may not always match the security maturity of larger markets. Safe internet awareness among users is as important as the technology itself. A technically robust authentication system offers

limited protection if users share OTPs, use weak PINs, or fail to recognize phishing attempts.

Nepal has seen this play out in practice. In early 2026, a fraud network was uncovered in which citizenship documents collected at remittance counters were sold to third parties, who used them to create verified but fraudulent digital wallet accounts. The scheme resulted in losses of nearly NPR 4 million before seven individuals were arrested [11]. This incident highlights a critical weakness in the KYC onboarding process rather than in the authentication technology itself. It also illustrates that security in Nepal's digital payment ecosystem cannot be evaluated in isolation from the broader social and procedural context in which these apps operate.

Encryption standards and blockchain-based transparency have been discussed as potential solutions for strengthening transaction integrity [5]. However, their deployment within Nepal's current infrastructure remains limited, and identity theft along with cybercrime continue to be identified as the most pressing barriers to digital payment trust [2].

### 2.4 Usability and User Experience

Security and usability represent competing design priorities in authentication systems. Additional authentication layers improve security but introduce user friction. A comparative study on single-factor versus two-factor authentication found that while 2FA is generally accepted as trustworthy, it introduces cognitive burden and additional steps that frustrate users, particularly those less familiar with digital systems [6]. Three dimensions define how users evaluate authentication usability: ease of use, required cognitive effort, and perceived trustworthiness.

In the Nepalese context, this tension takes on a local dimension. Trust in digital payment platforms is directly and strongly connected to perceived security and privacy [4]. At the same time, if a platform is not perceived as easy to use, users will not trust its overall quality regardless of how secure it actually is. This creates a dual requirement where platforms must be both genuinely secure and feel effortless to use, which is harder to achieve than either goal alone.

Interestingly, research on Nepali fintech users found that higher perceived risk did not always deter adoption. In some cases, it was associated with increased usage, suggesting that convenience outweighs security concerns for a segment of users [3].

This implies that some users are knowingly accepting risk rather than dealing with authentication friction. Session persistence features that reduce login steps are a practical example of this trade-off. They lower friction noticeably but increase vulnerability if a device is lost or stolen.

Biometric methods, when implemented well, offer a partial resolution to this tension. Multimodal systems have shown that security improvements do not have to come at the cost of user experience, maintaining authentication speed while significantly reducing error rates [9]. The challenge in Nepal is that the hardware required to support strong biometric authentication is not uniformly available, meaning usability benefits are unevenly distributed across the user population.

### 2.5 Research Gap

The existing literature covers digital payment security and usability from several angles. Global studies have examined authentication frameworks, biometric methods, and the general security-usability trade-off [6-10]. Nepal-focused studies have documented the country's digital payment landscape, consumer trust patterns, and fintech adoption behavior [2-4]. What is missing is a study that brings these threads together in a comparative, platform-specific analysis grounded in the Nepali context.

No published research has directly compared the authentication mechanisms of eSewa, Khalti, and a major commercial bank's mobile application side by side. Existing Nepal-focused studies tend to treat digital payments as a broad category rather than examining specific platforms and their individual implementation choices. eSewa and Khalti are fintech-native wallets with faster development cycles, while commercial bank apps operate under stricter regulatory constraints and more conservative security postures. Comparing them directly should reveal not just which is more secure or more usable, but why those differences exist and what can be done about them.

Additionally, the gap between what authentication technology exists globally and what is actually deployed in Nepal's market has not been systematically documented. The inconsistent support for face recognition due to hardware limitations, the continued reliance on SMS OTP despite its known vulnerabilities, and the complete absence of behavioral biometrics across all three platforms are areas where local reality diverges from global best practice. This study aims to document that divergence carefully and offer grounded, context-aware recommendations for closing it.

# 3. Research Methodology

## 3.1 Research Design

This study employs a mixed-methods design combining quantitative survey data with qualitative app analysis. The approach is descriptive and comparative. Rather than conducting controlled laboratory experiments, the goal is to examine real-world platforms to test the stated hypothesis and compare how three platforms currently handle authentication, and what users actually think about those mechanisms.

A comparative design is appropriate here because the study involves three distinct platforms that serve overlapping user populations but operate under different development models. eSewa and Khalti are fintech-native wallets, while the commercial bank app operates under stricter regulatory oversight. Placing them side by side makes it possible to identify not just what differs between them, but why those differences exist. Figure 1 illustrates the overall research framework.

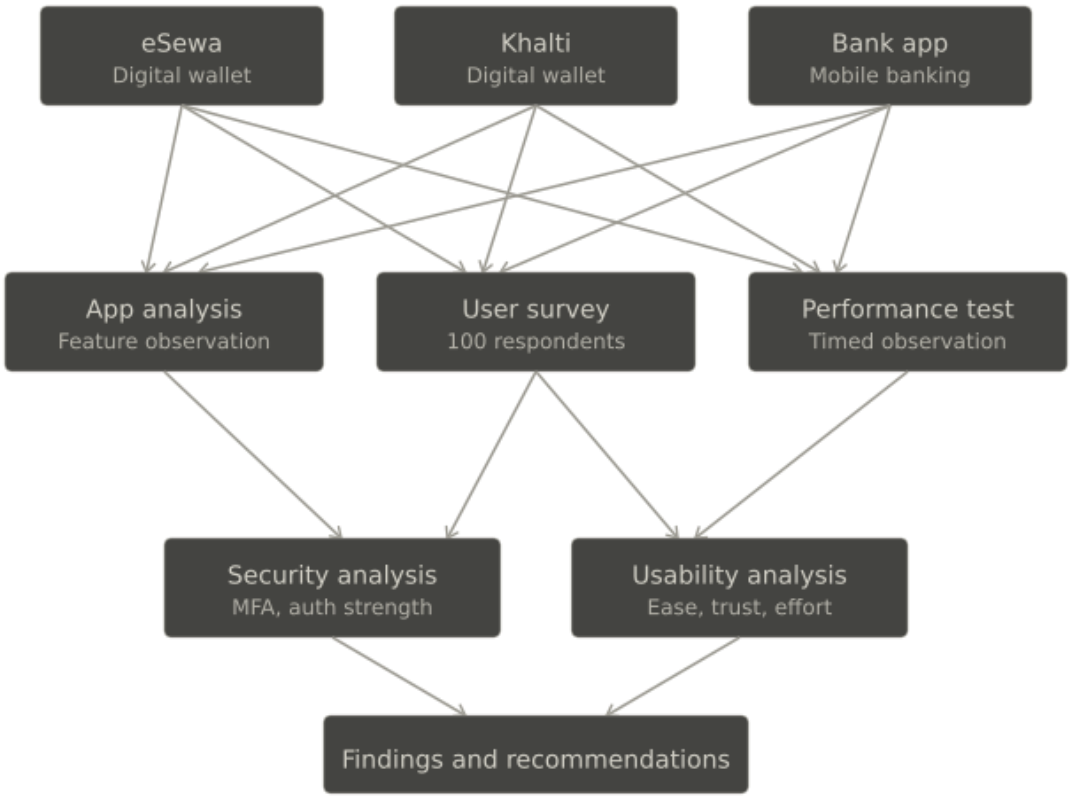


*Figure 1: Research Framework*

The scope of this research is focused on the front-end authentication layer, meaning what a user directly interacts with during login and transaction verification. This includes authentication methods such as PIN, biometrics, and OTP, user experience and usability metrics, and user perception of security and trust. Backend infrastructure, regulatory compliance, and international comparisons fall outside the scope of this study.

## 3.2 App Selection Criteria

Three platforms were selected based on their market relevance, diversity of platform type, and accessibility for direct testing. Table 1 summarizes the selection.

*Table 1: Selected Applications*

| App | Type | Market Position | Reason for Selection |
|---|---|---|---|
| eSewa | Digital wallet | Market leader | Largest user base in Nepal |
| Khalti | Digital wallet | Second largest | Primary competitor to eSewa |
| Major commercial bank app | Mobile banking | Established institution | Represents bank-integrated model |

eSewa and Khalti together represent the dominant share of Nepal's digital wallet market, making them the natural starting point for any study on Nepali digital payment security. Including a commercial bank app alongside them is important because bank apps operate under Nepal Rastra Bank regulations and tend to take a more conservative approach to feature rollout, particularly around biometrics. Comparing these two platform types reveals how the regulatory environment shapes authentication design in practice.

All three apps are freely available for download and can be tested by any registered user without special access. This makes hands-on analysis feasible without requiring cooperation from the companies themselves.

## 3.3 Data Collection Methods

Data for this study will be collected through three complementary methods.

Method 1: App Analysis

Each app will be installed and tested across different devices to capture variation in biometric authentication support, as discussed in the literature review.

The analysis will document the full authentication flow for each platform, covering initial login and transaction verification steps. A standardized checklist will be used across all three apps to ensure consistency.

*Table 2: Authentication Feature Comparison*

| Feature | eSewa | Khalti | Major bank app |
|---|---|---|---|
| MPIN / Password | Yes | Yes | Yes |
| Biometric login (fingerprint / face) | Device dependent | Device dependent | Device dependent |
| SMS OTP | Yes | Yes | Yes |
| Behavioral biometrics | No | No | No |

Beyond listing features, the analysis will also note usability aspects such as how many steps each authentication method requires, whether error messages are clear and helpful, and whether the app provides any guidance to users about their security settings.

Method 2: User Survey

A structured survey will be administered to approximately 100 respondents. The sample is intentionally diverse, drawing from urban internet users, people at nearby shops and local businesses, and residents from remote and village areas. This diversity is important for generalizability. A study that only asks urban, tech-comfortable people wouldn't represent all the different types of people actually using these apps across Nepal.

The survey will cover three areas.

i. Basic usage patterns: which apps respondents use, how frequently, and for what purposes.

ii. Security awareness: whether respondents know what authentication methods their app uses, whether they have enabled biometrics, and whether they have experienced any security incidents.

iii. Usability perception: how easy or difficult respondents find the login process, whether they feel the security steps are appropriate, and how much they trust their chosen platform.

Survey design draws on established usability dimensions from the literature, specifically ease of use, cognitive effort, and perceived trustworthiness [6].

Method 3: Performance Observation

This method involves timed observation of authentication processes across the three apps. Each login method, PIN entry, biometric scan, and OTP verification, will be tested multiple times and the time taken recorded. Authentication success and failure rates under normal conditions will also be noted. This produces a basic quantitative picture of how each method performs in practice, complementing the qualitative observations from Method 1 and the perception data from Method 2.

### 3.4 Data Analysis Approach

Survey data will be analyzed using descriptive statistics. Responses will be summarized as frequencies and percentages to show patterns in usage, awareness, and perception across the sample. Where relevant, responses will be broken down by user group, for example comparing urban versus rural respondents, or eSewa users versus Khalti users, to identify meaningful differences.

App analysis data will be organized into a comparative framework based on two dimensions drawn from the literature. The first is security strength, evaluated against MFA standards and authentication categories identified in prior research [7][8]. The second is usability, evaluated against the three dimensions of ease of use, cognitive effort, and trustworthiness [6]. Each platform will be assessed on both dimensions, allowing a side-by-side comparison that goes beyond simply listing features. The analysis will also evaluate whether findings support or challenge H1.

Performance observation data will be presented as simple averages and compared across platforms and authentication methods. This will help illustrate where authentication friction is highest and where improvements in speed or reliability could improve user experience without compromising security.

### 3.5 Ethical Considerations

This study involves human participants through the user survey and therefore requires careful attention to ethical standards. All survey participation will be fully voluntary. Respondents will be informed of the purpose of the study before they agree to participate, and they may withdraw at any time without consequence. No personally

identifiable information will be collected. Responses will be recorded anonymously and stored securely.

The app analysis component involves only publicly available applications tested through normal user accounts. No attempt will be made to access backend systems, bypass security controls, or test for vulnerabilities through unauthorized means. The study observes and documents existing behavior rather than probing for weaknesses through intrusive methods.

Data collected through the survey will be used solely for the purposes of this research and will not be shared with third parties, including the companies whose apps are being studied. Given that the study touches on security awareness and personal financial habits, maintaining respondent confidentiality is particularly important.

### 3.6 Limitations

This study has several limitations. First, this study relies on publicly observable app behavior and user-reported perceptions. Internal security architecture and backend fraud detection systems are outside the scope of what can be analyzed here. The findings therefore reflect the user-facing layer of each platform rather than its complete security posture.

Second, a survey sample of around 100 respondents, while diverse in composition, is modest in size. Prior Nepal-based studies have used larger samples to draw stronger generalizations. The findings of this study should be interpreted as indicative rather than definitive, and future research with a larger sample could strengthen the conclusions.

Third, app behavior varies across devices and operating system versions. While testing across multiple devices captures hardware variation, it cannot fully represent the complete range of devices used across Nepal.

Fourth, survey responses are self-reported and subject to individual bias. Respondents may overestimate their own security awareness or underreport security incidents. This is a common limitation in usability and security perception research [6].

# 4. Expected Outcomes

## 4.1 Anticipated Findings

Based on the literature reviewed and preliminary observation of the three platforms, this study anticipates several findings. First, the app analysis is expected to confirm that all three platforms rely heavily on knowledge-based authentication, specifically MPINs and passwords, as their primary login mechanism, with biometric login available but dependent on user device capability. Behavioral biometrics and continuous authentication are expected to be entirely absent across all three platforms.

Second, the survey data is expected to reveal that a significant portion of Nepali users, particularly those from rural and semi-urban areas, are unaware of what authentication methods their apps actually use or whether they have enabled available security features like biometric login. This would reflect a broader pattern visible in Nepal's digital adoption landscape.

Third, in terms of H1, the findings are expected to support the claim that digital wallet apps prioritize usability over security. Features like simplified login flows in wallet apps contrast with the stricter, more step-heavy authentication processes typically found in bank apps. The performance observation is likely to show that wallet apps authenticate faster but with fewer security layers, while bank apps are slower but more thorough.

Fourth, user perception data is expected to show a strong correlation between trust and perceived security, a pattern that mirrors what Nepal's digital payment landscape broadly suggests. Users who feel their app is secure are likely to rate it more positively overall, regardless of actual security implementation, which points to the importance of security communication alongside security design.

## 4.2 Contributions and Recommendations

This study contributes to the existing body of knowledge in two ways. First, it fills a specific gap by providing the first direct comparison of authentication mechanisms across eSewa, Khalti, and a commercial bank app in Nepal, grounded in both technical observation and user perception data. Second, it grounds global authentication research in a local context, showing where Nepal's platforms align with or diverge from international best practice.

In terms of practical recommendations, this study anticipates recommending the following directions. App developers should consider strengthening OTP-based verification by moving toward app-based authentication rather than SMS-only OTP, which carries known vulnerabilities. Platforms should also invest in clearer in-app communication about available security features, since many users may not know these options exist. For policymakers and Nepal Rastra Bank, the findings could inform updated guidelines around minimum authentication standards for digital wallets, which currently appear less stringent than those applied to bank apps.

Ultimately, improving digital payment security in Nepal does not require dramatic technological leaps. Small, targeted improvements in authentication design, combined with better user awareness, could meaningfully strengthen the ecosystem while keeping the ease of use that drives adoption in the first place.

## References


[1] Laxmi Bank Limited, "Laxmi Bank Limited Annual Report 2003/2004," 2004. [Online]. Available: https://www.laxmisunrise.com/wp-content/uploads/2022/06/2003-04-english.pdf

[2] A. Pathak, "Digital Payment In Nepal: An Overview And Recommendations," Rupandehi Campus J., vol. 4, no. 1, pp. 25–34, Feb. 2024, doi: 10.3126/rcj.v4i1.62916.

[3] A. Nepal, "The Impact of Fintech Innovations on Traditional Banking Practices and Customer Preferences in Nepal," Nep. J. Finance, vol. 12, no. 2, pp. 74–87, Nov. 2025, doi: 10.3126/njf.v12i2.83114.

[4] O. Poudel and M. P. Sapkota, "Consumer Perception toward Digital Payment System," Management Dynamics, vol. 25, no. 1, pp. 39–50, Dec. 2022, doi: 10.3126/md.v25i1.53286.

[5] Md. A. Hassan and Z. Shukur, "A Systematic Review of User Authentication Security in Electronic Payment System," in Proceedings of International Conference on Data Science and Applications, vol. 551, M. Saraswat, C. Chowdhury, C. Kumar Mandal, and A. H. Gandomi, Eds., in Lecture Notes in Networks and Systems, vol. 551., Singapore: Springer Nature Singapore, 2023, pp. 121–138. doi: 10.1007/978-981-19-6631-6_10.

[6] G. M. Khaskheli, M. Sherbaz, and U. R. Shaikh, "A comparative usability study of single-factor and two-factor authentication," Tropical Scientific Journal, vol. 1, no. 1, pp. 17–27, Feb. 2022.

[7] N. A. Karim, O. A. Khashan, H. Kanaker, W. K. Abdulraheem, M. Alshinwan, and A.-K. Al-Banna, "Online Banking User Authentication Methods: A Systematic Literature Review," IEEE Access, vol. 12, pp. 741–757, 2024, doi: 10.1109/ACCESS.2023.3346045.

[8] H. U. Khan, M. Sohail, S. Nazir, T. Hussain, B. Shah, and F. Ali, "Role of authentication factors in Fin-tech mobile transaction security," J Big Data, vol. 10, no. 1, p. 138, Sep. 2023, doi: 10.1186/s40537-023-00807-3.

[9] K. Al-Fatlawi, M. M. A. Zaid, L. S. Alyassri, A. H. Al-Fatlawi, A. A. H. Al-Magsoosi, and O. M. Albayati, "Biometric Authentication Methods for Strengthening Access Control in Wallets and Digital Finance Applications," in 2025 3rd International Conference on Cyber Resilience (ICCR), Dubai, United Arab Emirates: IEEE, Jul. 2025, pp. 1–6. doi: 10.1109/ICCR67387.2025.11291664.

[10] S. Riyana, "Enhancing Security in Digital Wallets Using Multi-Factor Authentication and Behavioral Biometrics," in IJETCSIT, May 2025, pp. 559–570. doi: 10.56472/ICCSAIML25-162.

[11] Techpana, "IME Remittance Counter Users' Citizenship Misused For Online Fraud, Seven Arrested," Techpana. Accessed: Jan. 06, 2026. [Online]. Available: https://techpana.com/2026/154818/ime-remittance-counter-user-citizenship-misused-for-online-fraud-seven-arrested